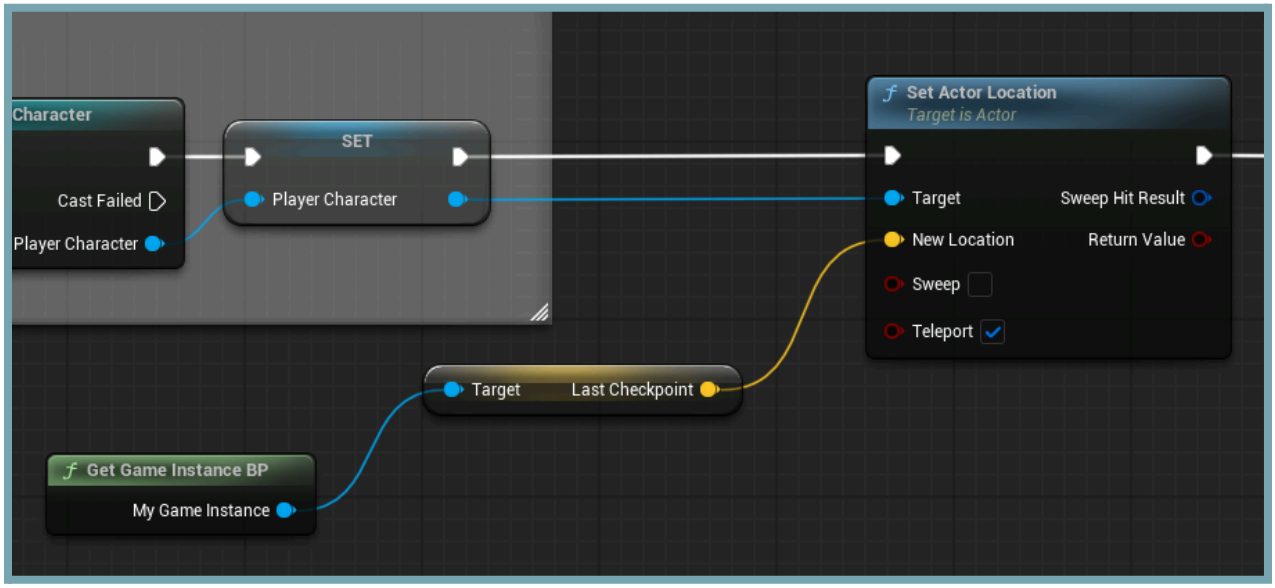
CloudWorks Casino – Debugging Comments

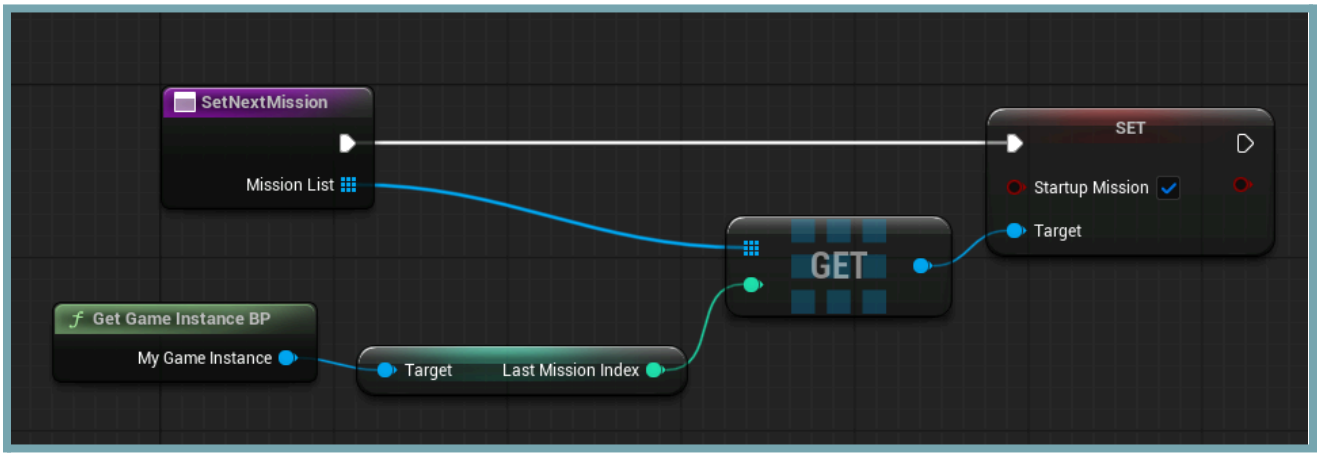
New Functionality

BP_Checkpoint + BP_Mission_Holder

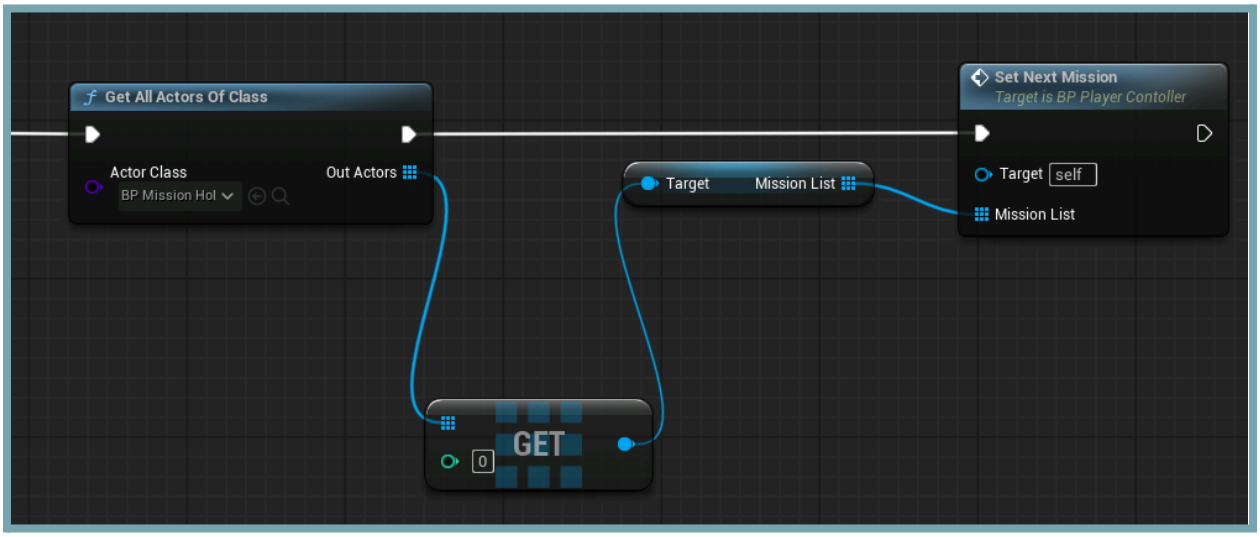
BP_Checkpoint Path: Content\Blueprints\Gameplay
BP_Mission_Holder Path: Content\Blueprints\Missions\Items

- Checkpoints save a respawn location for the player as well as the current mission they should have at that point.
- Used by the Game Instance BP to remain persistent between level loading.
- Player Controller BP now initializes with the respawn point and mission index from the Game Instance BP





- The Mission Holder is used to access a list of the missions between level loading. Since the game instance cannot access the mission master objects between level loads, the game instance finds the mission holder object when the game starts and sets the startup mission based on the index in the list.



- Checkbox added to Main Menu to allow the player to choose if they would like checkpoints to be active or not.



BP_End + WB_MissionAccomplished

BP_End Path: Content\Blueprints\Gameplay
WB_MissionAccomplished Path: Content\UI\Widgets

- Simple trigger box which activates the Mission Accomplished widget, telling the player they completed the level.
- Widget has one button which takes the player back to the main menu.



BP_EscapeSequence + WB_EscapeTimer

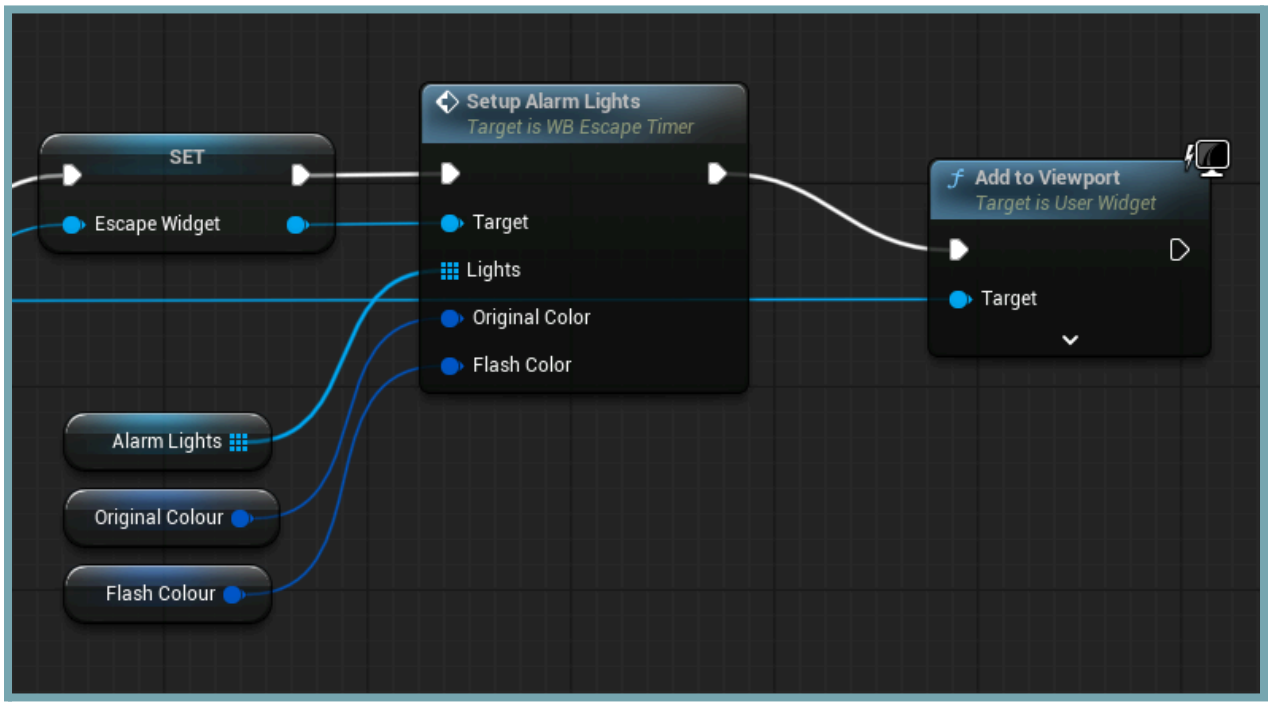
BP_EscapeSequence Path: Content\Blueprints\Gameplay
WB_EscapeTimer Path: Content\UI\Widgets

- Escape Sequence is another simple trigger box which activates the Escape Timer widget.
- The Escape Timer widget starts with a text announcement telling the player that the escape sequence has begun. After 5 seconds of being active, the announcement will be replaced by a one-minute countdown timer.



- The Escape Timer has a list of Basic Light objects. Every other second, the Escape Timer will take each of these lights and will turn it either red or white to simulate a slow flashing effect for the escape sequence.
- The Escape Sequence BP sends a list of Basic Light objects, as well as the original colour and flash colour (white and red) that the Escape Timer WB will use.





Power System Changes

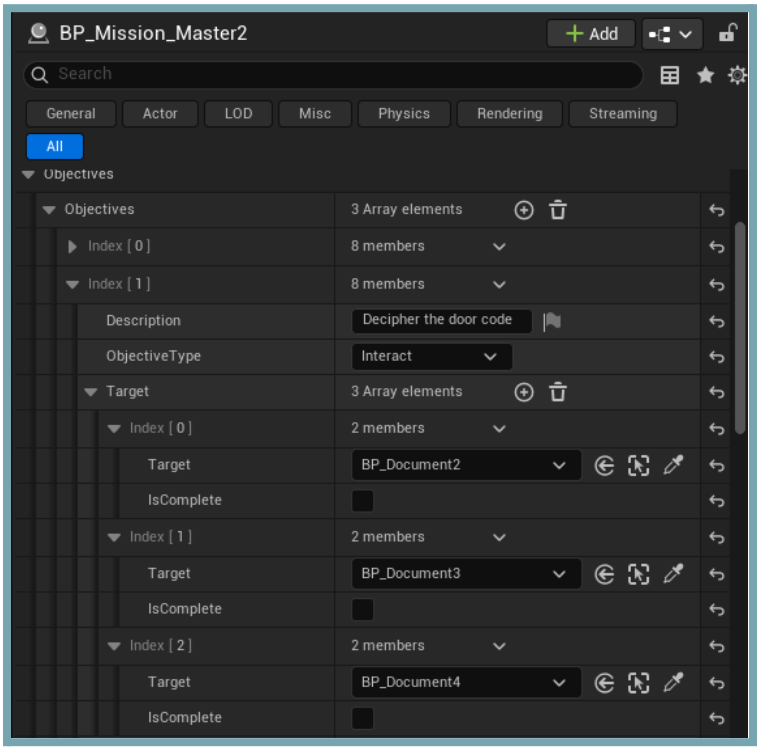
- Fixed breakers and switches to switch the power state of the system.
- Added events to lasers and cameras to listen for breakers changing the power state.
- Edited lasers to kill players to give them a larger threat aspect and stop players from simply running past them.

Missions & Objectives

- Allowed the Door BP and Power Breaker BP to call the parent Interact function to allow for them to activate objectives.

Multiple Objective Targets

- Added **S_ObjectiveTarget** structure to keep track of a singular target of an object. (Path: Content\Blueprints\Missions\Data\Structs)
- Allow multiple targets (of type S_ObjectiveTarget) to be needed for one objective. The functions *CheckLocationObjective* and *CheckInteractObjective* on Mission Master BP now check for the actor on the S_ObjectiveTarget list of each S_ObjectiveData, and if the objective target has been found and interacted with, the target will set its "IsCompleted" variable to true.
- The Mission Master now has a *CheckObjectiveComplete* function which will go through an objective's list of targets and see if they are all complete or not. The objective is only complete once all of its targets have been completed. This is used in the second objective of the third mission in the level.



Small Tweaks

- Added document text to VIP to be able to talk to them.
- Decreased sounds from player running and falling to make gameplay feel more consistent as guards could hear from too far away which would make gameplay feel unpolished.
- Reduced camera view length as they could see through floors.
- Gave locked doors an “CanInteract” variable so that they can be used as set dressing without confusing the player by displaying interact text or playing interact sounds.
- Modified elevator to use Lerp (Vector) rather than VInterp To since the latter was causing its stopping points to be determined by the frame rate of the device running the build. This means that it would at times stop before actually reaching the floor, leaving a small gap for the player.